

EEE102 – Elements of Electrical Engineering

Department of Electrical and Electronics Engineering NMAM Institute of
Technology Nitte,

Karkala- 574110 Odd Semester 2025-26

Unit- I: Graph Theory in Electrical Circuits

Contents

1.1	Introduction	
1.2	Ohm's law	
1.3	Kirchhoff Laws (KVL, KCL)	
1.4	Power	
1.5	Graph Representations	
1.5.1	Terminology used in the network graph	
	Node	
	Branch	
	Loop	
	Mesh	
1.5.2	Linear graph of a network	
1.5.3	Tree	
1.5.4	Incidence Matrix	
1.5.5	Tie set	
1.5.6	Cut-set Matrix	

1.1 Introduction

The interconnection of various electric elements in a prescribed manner comprises as an electric circuit in order to perform a desired function. The electric elements include controlled and uncontrolled source of energy, resistors, capacitors, inductors, etc. Analysis of electric circuits refers to computations required to determine the unknown quantities such as voltage, current and power associated with one or more elements in the circuit. To contribute to the solution of engineering problems one must acquire the basic knowledge of electric circuit analysis and laws. In electrical engineering, solution of network analysis problems involves finding the current through and voltage across different circuit elements. The graph theory deals with the properties of networks which do not change with the change in the shape of the networks. The graph theory concept eases the solution method for solving networks with a large number of nodes and branches.

1.2 Ohm's law

One of the most important steps in the analysis of the circuit was undertaken by Georg Ohm, who found that the potential difference across the ends of many conductors is proportional to the current flowing between them. This, he found, was a direct proportionality, provided that temperature remained constant. The mathematical expression of Ohm's law is given as follows:

$$V \propto I$$

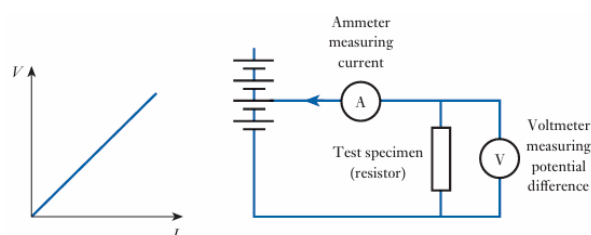


Fig. 1. Constant potential difference/current characteristic and the circuit from which it is obtained.

$$\frac{V}{I} = R$$

Where R is a constant termed the resistance of the conductor. The expression involving R is usually expressed as

$$V = IR$$

Ohm's law only notes the constancy of potential difference to current provided that other physical factors remain unchanged.

1.3 Kirchhoff Laws

Gustav Kirchhoff, a German physicist, observed that these were instances of two general conditions fundamental to the analysis of any electrical network. These conditions may be stated as follows:

1.3.1 Kirchhoff's Current Law

Kirchhoff's Current Law (KCL) states that the total current entering a node (junction) in an electrical circuit is equal to the total current leaving that node. This is also known as Kirchhoff's first law or the junction rule. Essentially, it reflects the principle of conservation of charge, meaning that charge is not lost or gained within a circuit.

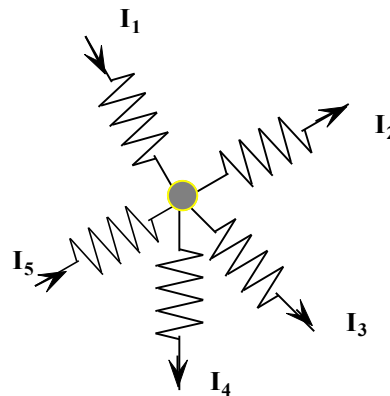


Fig. 2. Electric Circuit

Convention: Current entering a node + ve, leaving the node - ve

$$I_1 - I_2 - I_3 - I_4 + I_5 = 0$$

$$\text{i.e. } I_1 + I_5 = I_2 + I_3 + I_4$$

1.3.2 Kirchhoff's Voltage Law

Kirchhoff's Voltage Law (KVL) states that the sum of all voltage drops around any closed loop

in a circuit is equal to zero. This law is a consequence of energy conservation in electrical circuits. In simpler terms, the total voltage source (like a battery) in a loop must equal the total voltage drops across the components in that loop. Consider the closed electric circuit shown in Fig 3.

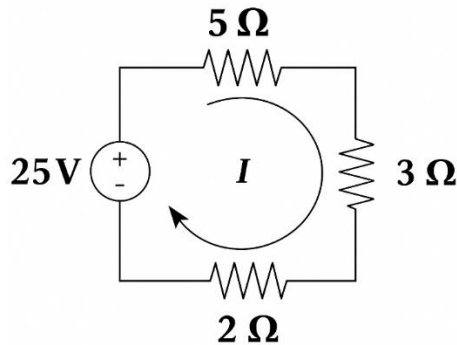


Fig 3. Electric Circuit

Convention: Voltage drops -ve. Source – to + is + ve and + to – is -ve

Hence applying to KVL to the circuit shown in Fig.3.

$$-5I - 3I - 2I + 25 = 0$$

$$10I = 25$$

$$I = 2.5 \text{ A}$$

1.4 Power

Power is an indication of how much work (the conversion of energy from one form to another) can be done in a specified amount of time, that is, a rate of doing work. The electrical unit of measurement for power is the watt (W), defined by 1 watt (W) = 1 joule/second (J/s).

$$P = VI = V \left(\frac{V}{R} \right) = \frac{V^2}{R} \text{ W}$$

$$P = VI = (IR)I = I^2 R \text{ W}$$

Example:

1. Apply KVL for the network shown in Fig. 4, find the power absorbed by 18Ω resistor and power supplied by 10V voltage source.

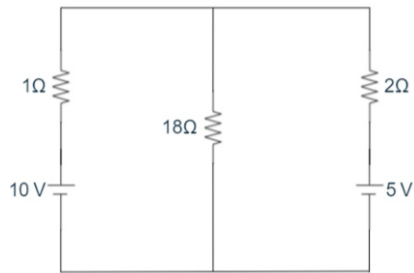
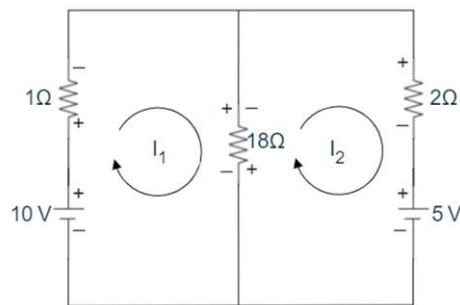


Fig 4. Electric Circuit

Solution:



Loop 1: $10 - 1(I_1) - 18(I_1 - I_2) = 0$

Loop 2: $-18(I_2 - I_1) - 2(I_2) - 5 = 0$

$$10 - 1(I_1) - 18(I_1) + 18(I_2) = 0 \implies -19I_1 + 18I_2 = -10$$

$$-18(I_2) + 18(I_1) - 2(I_2) - 5 = 0 \implies 18I_1 - 20I_2 = 5$$

$$I_1 = 1.9642A \text{ and}$$

$$I_2 = 1.5178A$$

The current through the 10V source and 1 Ω resistor is 1.9642A and the current through the 5V source and 2 Ω resistor is 1.5178A. The current through the 18 Ω resistor is determined by the following equation from the original network:

$$I_{18\Omega} = I_1 - I_2 = 1.9642 - 1.5178 = 0.4464A(\downarrow)$$

$$\text{Power absorbed by } 18\Omega = I^2 \times 18 = 3.58691W$$

$$\text{Power supplied by } 10\text{ V battery} = 1.9642 \times 10 = 19.642W$$

2. Apply KCL for the network shown in Fig. 5 to calculate the voltages V_1 and V_2 .

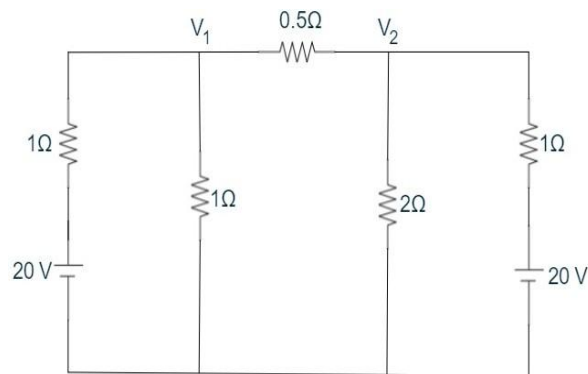
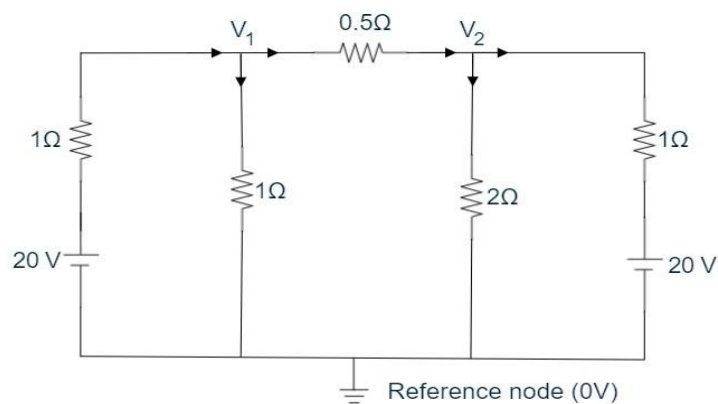


Fig 5. Electric Circuit

Solution:



At node 1:

$$\left(\frac{V_1 - 0}{1}\right) + \left(\frac{V_1 - V_2}{0.5}\right) = \left(\frac{20 - V_1}{1}\right)$$

$$4V_1 - 2V_2 = 20 \quad (1)$$

At node 2:

$$\left(\frac{V_1 - V_2}{0.5}\right) = \left(\frac{V_2 - 20}{1}\right) + \frac{V_2 - 0}{2}$$

$$2V_1 - 3.5V_2 = -20 \quad (2)$$

Solving Eq. (1) and (2) gives $V_1 = 11$ V and $V_2 = 12$ V

2.5 Graph representation

The graph of a network is drawn by first marking the nodes and then joining these nodes by lines which correspond to the network elements of each branch. All the voltage and current sources are replaced by their internal impedances. The voltage sources are replaced by short

circuits as their internal impedances are zero whereas current sources are replaced by open circuits as their internal impedances are infinite.

1.5.1 Terminology used in the network graph

Node

It is defined as a point at which two or more elements have a common connection.

Example: a, b, c, d, e, f and g

Branch

It is a line connecting a pair of nodes, the line representing a single element or series connected elements.

Example: v_1 , R_1 , R_2 , R_3 , v_2 , R_4 , R_5 , R_6 , R_7 and I

Loop

It is the closed contour selected in a graph.

Example: abedca, abegfca, cdebgfc

Mesh

It is a loop which does not contain any other loops within it.

Example: abedca, cdebgfc, gebg (through R_7) and gebg (through I)

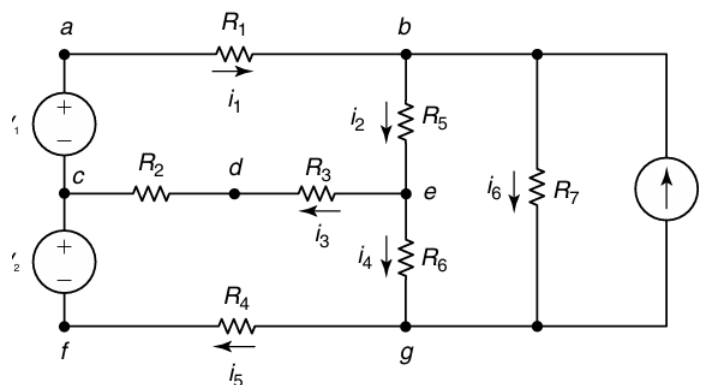


Fig 5. Circuit illustrating terminologies

1.5.2 Linear graph of a network

A linear graph is a collection of nodes and branches. The nodes are joined together by branches.

Figure 5 shows a network and its associated graphs. Each branch of a graph may be given an orientation or a direction with the help of an arrow head which represents the assigned reference direction for current. Such a graph is then referred to as a directed or oriented graph. Branches whose ends fall on a node are said to be incident at that node. Degree of a node is defined as

the number of branches incident to it. Branches 2, 3 and 4 are incident at Node 2 in Fig. 6 (c). Hence, the degree of Node 2 is 3.

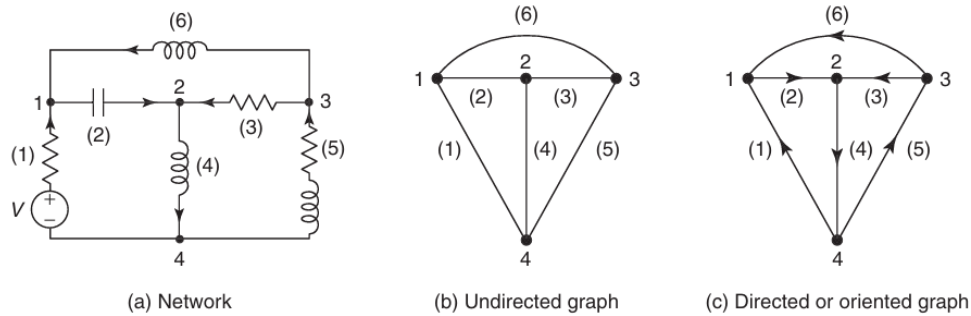


Fig 6. Network and its graphs

1.5.3 Tree

It is an interconnected open set of branches which include all the nodes of the given graph. In a tree of the graph there cannot be any closed loop.

Tree branch or Twig

It is any branch of a tree. The number of twigs on a tree is always one less than the number of nodes.

Let N = Number of nodes; then number of twigs = $N - 1$

Tree link or chord

It is that branch of the graph that does not belong to the particular tree.

The number of links (L) = $B - N + 1$

Where,

B = Number of branches

N = Number of nodes

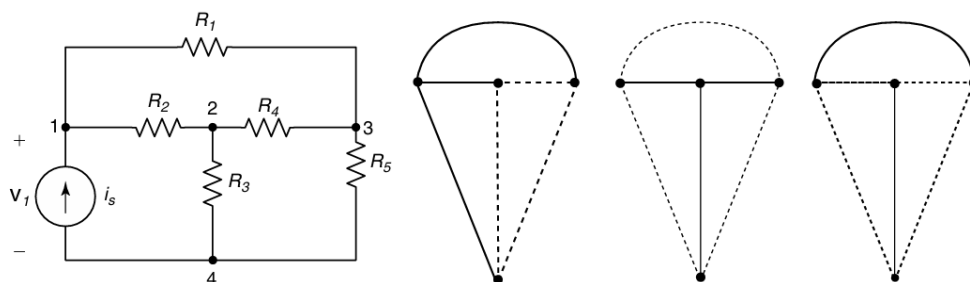


Fig 7. Circuit and possible trees of the circuit

1.5.4 Incidence matrix $[A_a]$

The incidence matrix symbolically describes a network. It also facilitates the testing and identification of the independent variables. The incidence matrix which represents a graph uniquely. This matrix tells us which branches are incident at which nodes and what the orientations relative to the nodes are.

For a given graph with 'n' nodes and 'b' branches, the complete incidence matrix is a rectangular matrix of order $n \times b$, whose elements have the following rules:

Number of columns in $[A_a] = \text{Number of branches} = b$

Number of rows in $[A_a] = \text{Number of nodes} = n$

The matrix obtained from incidence matrix by eliminating one of the rows is called the reduced incidence matrix. In other words, suppression of the datum node (reference node) from the incidence matrix results in a reduced incidence matrix $[A]$.

Properties of Incidence matrix $[A_a]$

- (i) The sum of the entries in any column is zero.
- (ii) The determinant of the incidence matrix of a closed loop is zero.
- (iii) The rank of the incidence matrix of a connected graph is $(n - 1)$.

Number of possible trees of a graph

The number of possible trees of a graph, $= \det \{[A] \times [A]^T\}$

where, A is the reduced incidence matrix obtained by eliminating any one row of the complete incidence matrix A_a , and $[A]^T$ is the transpose of the matrix $[A]$.

1.5.5 Tie-set

It is a unique set with respect to a given tree of a connected graph containing one chord and all of the free branches contained in the free path formed between two vertices of the chord. Consider the graph and the tree as shown in Fig. 8. The selected tree will result in three fundamental loops as we connect each link, in turn to the tree.

For a given graph having 'n' nodes and 'b' branches, the **tie-set matrix** is a rectangular matrix with 'b' columns and as many rows as there are loops.

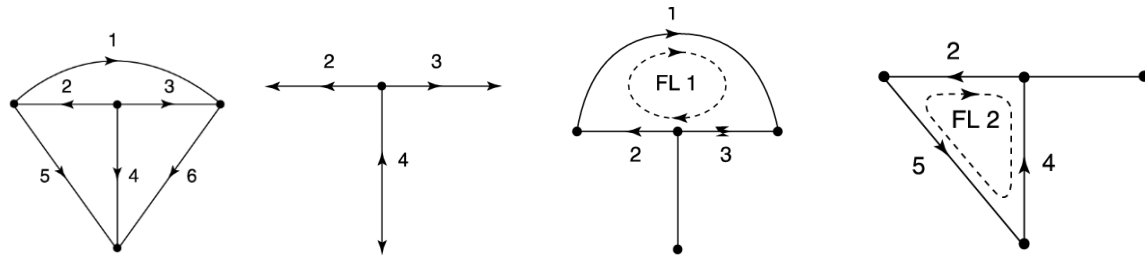
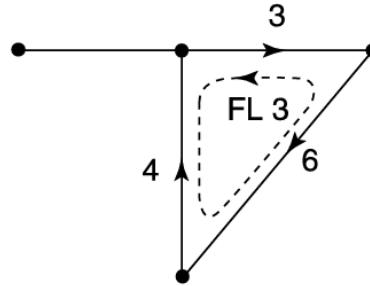


Fig 8. (a) Graph

(b) Tree of the graph

(c) Loop 1

(d) Loop 2



(e) Loop 3

Fundamental Loop 1 (FL1): Connecting link 1 to the tree

Fundamental Loop 2 (FL2): Connecting link 5 to the tree

Fundamental Loop 3 (FL3): Connecting link 6 to the tree

These sets of branches (1, 2, 3), (2, 4, 5) and (3, 4, 6) form three tie-sets.

1.5.6 Cut-set

It is that set of elements or branches of a graph that separates two main parts of a network. If any branch of the cut-set is not removed, the network remains connected. The term cut-set is derived from the property designated by the way by which the network can be divided into two parts. A cut-set is shown on a graph by a dashed line or solid line which passes through the branches defining the cut-set. A graph should have at least one cut set though there may be more than one cut-set in any graph. A fundamental cut-set is a cut-set that cuts or contains one and only one tree branch. Therefore, for a given tree the number of fundamental cut-sets will be equal to the number of twigs. For a given graph, a **cut-set matrix** $[Q_a]$ is defined as a rectangular matrix whose rows correspond to cut-sets and columns corresponds to the branches of the graph.

Procedure for finding the fundamental cut-set

1. First select a tree of the given graph.
2. Focus on a tree branch.

3. Check whether removing the tree branch from the tree disconnects the tree into two separate parts.
4. All the links which go from one part of the disconnected tree to the other, together with the tree branch forms a fundamental cut-set.

Properties of a cut-set

1. A cut-set divides the set of nodes into two subsets.
2. Each fundamental cut-set contains one tree-branch, the remaining elements being links.
3. Each branch of the cut-set has one of its terminals incident at a node in one subset and its other terminal at a node in the other subset.
4. A cut-set is oriented by selecting an orientation from one of the two parts to the other. Generally, the direction of a cut-set is chosen same as the direction of the tree branch.